

TUTORIAL 02 – THE DATA LEAKAGE CASE

Scenario

‘Iaman Informant’ was working as a manager of the technology development division at a famous international company OOO that developed state-of-the-art technologies and gadgets.

One day, at a place which ‘Mr. Informant’ visited on business, he received an offer from ‘Spy Conspirator’ to leak of sensitive information related to the newest technology. Actually, ‘Mr. Conspirator’ was an employee of a rival company, and ‘Mr. Informant’ decided to accept the offer for large amounts of money, and began establishing a detailed leakage plan.

‘Mr. Informant’ made a deliberate effort to hide the leakage plan. He discussed it with ‘Mr. Conspirator’ using an e-mail service like a business relationship. He also sent samples of confidential information though personal cloud storage.

After receiving the sample data, ‘Mr. Conspirator’ asked for the direct delivery of storage devices that stored the remaining (large amounts of) data. Eventually, ‘Mr. Informant’ tried to take his storage devices away, but he and his devices were detected at the security checkpoint of the company. And he was suspected of leaking the company data.

At the security checkpoint, although his devices (a USB memory stick and a CD) were briefly checked (protected with portable write blockers), there was no evidence of any leakage. And then, they were immediately transferred to the digital forensics’ laboratory for further analysis.

The information security policies in the company include the following:

- (1) Confidential electronic files¹ should be stored and kept in the authorized external storage devices and the secured network drives.
- (2) Confidential paper documents and electronic files can be accessed only within the allowed time range from 10:00 AM to 16:00 PM with the appropriate permissions.
- (3) Non-authorized electronic devices such as laptops, portable storages, and smart devices cannot be carried onto the company.
- (4) All employees are required to pass through the ‘Security Checkpoint’ system.
- (5) All storage devices such as HDD, SSD, USB memory stick, and CD/DVD are forbidden under the ‘Security Checkpoint’ rules.

In addition, although the company managed separate internal and external networks and used DRM (Digital Rights Management) / DLP (Data Loss Prevention) solutions for their information security, ‘Mr. Informant’ had sufficient authority to bypass them. He was also very interested in IT (Information Technology), and had a slight knowledge of digital forensics.

In this scenario, find any evidence of the data leakage, and any data that might have been generated from the suspect’s electronic devices.

Task 01:

Calculate and verify the hash functions of the different images to ensure the integrity of the data.

Task 02:

Provide detailed information of the target systems and devices involved in the investigations. This information includes but not limited to the type of device, Manufacturer, File system, Volume label, Serial number, model, IP address etc., which can be used to identify these devices in the investigation.

Task 03:

To better understand the behavior and activities of the suspect, you should analyze the images and develop a detailed timeline, actions and descriptions of the different activities of the suspect, during the suspected period (A graphical timeline of his activities will be a plus).

QnA:

- 1 - What forensic branches in your opinion will be required to perform a detailed analysis of the suspects events? Defend your answers.
- 2 – Can you identify the partition information of PC image?
- 3 – What OS was running the suspect's computer? Explain your reasoning in details.
- 4 – Who is the owner of the computer? Explain how you got this information.
- 5 - List all accounts in OS except the system accounts.
- 6 - Who was the last user to logon into PC?
- 7 - Explain the information of network interface(s). Were the IPs assigned manually or automatically (DHCP) ?
- 8 - List application execution logs.
- 9 - List all traces about the system on/off and the user logon/logoff.
- 10 - What web browsers were used?
- 11 - Identify directory/file paths related to the web browser history.
- 12 - What websites were the suspect accessing? (Timestamp, URL...) ?
- 13 - List all search keywords using web browsers. (Timestamp, URL, keyword...)
- 14 - List all user keywords at the search bar in Windows Explorer. (Timestamp, Keyword)
- 15 - What application was used for e-mail communication?
- 16 - Where is the e-mail file located?
- 17 - What was the e-mail account used by the suspect?
- 18 - List all e-mails of the suspect. If possible, identify deleted e-mails.

- 19 - List external storage devices attached to PC.
- 20 - Identify all traces related to 'renaming' of files in Windows Desktop.
- 21 - List all directories that were traversed in the company's network drive.
- 22 - List all files that were opened in the company's network drive.
- 23 - Find traces related to cloud services on PC.
- 24 - What files were deleted from Google Drive?
- 25 - Identify account information for synchronizing Google Drive.
- 26 - List the e-mail communication stored in Windows Search database.
- 27 - List files and directories related to Windows Desktop stored in Windows Search database.
- 28 - What actions were performed for anti-forensics on PC at the last day '2015-03-25'?
- 29 - What hidden files were recovered from RM#3?

Based on your analysis and answers to the above questions, what evidence or proof do you have to support the fact that the suspect is guilty or not guilty of data leakage? Explain your reasoning and provide a report.